

Benchmarking Transformers and Baselines for Multi-Horizon Stock Return Prediction

Technical and Earnings Features on the Dow 30

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Key Question

Do complex transformer models outperform simpler baselines on daily stock data?

Why is this hard?

- Noisy, low signal-to-noise ratio
- Regime-dependent patterns
- Market efficiency limits

Our Contribution

- Fair comparison: identical features across all models
- Strict temporal validation
- Multiple horizons (1, 5, 21 days)
- Regime & event analysis
- Dow 30, 2016–2024

Three Core Hypotheses

H1: Model Class Advantage

Hypothesis: Temporal Fusion Transformer (TFT) will outperform baselines at $h = 21$ days

Rationale: Attention mechanisms should handle long-term dependencies better

Criteria: ≥ 2 pp improvement in DA or $\geq 5\%$ reduction in RMSE

H2: Event Features Add Value

Hypothesis: Adding earnings features improves DA without worsening RMSE

Rationale: Post-earnings-announcement drift is well-documented

Criteria: DA $\uparrow$ by ≥ 2 pp while RMSE stable

H3: Regime Dependence

Hypothesis: Model rankings differ between bear and bull markets

Rationale: Adaptive Markets Hypothesis suggests pattern instability

Criteria: Different best-performing models across 2022 bear vs. 2023–24 rally

Existing Approaches

- **Classical:** ARIMA, VAR—assume linearity/stationarity
- **Tree ensembles:** XGBoost, Random Forest—strong on cross-sections
- **RNNs:** LSTM/GRU capture temporal patterns
- **Transformers:** TFT, PatchTST—excel with scale & multimodal data
- **Recent:** TimesFM (zero-shot), BloombergGPT (text+price)

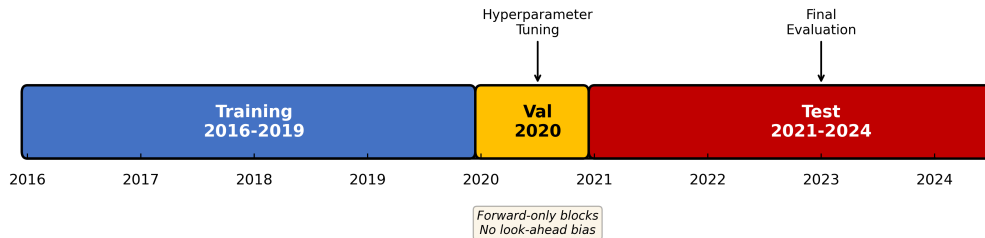
Key Gaps

- Few apples-to-apples comparisons across model classes
- Earnings/event features underused in deep models
- Look-ahead bias risks in evaluation
- Limited regime analysis

Our Contribution

- Fair comparison: identical features
- Earnings + technical inputs
- Strict temporal validation
- Regime & event breakdown

Experimental Design Overview



Data: Dow 30 stocks (fixed as of 2018)

Timeline:

- **Train:** 2016–2019 (4 years)
- **Validation:** 2020 (COVID year, held out)
- **Test:** 2021–2024 (4 years, multiple regimes)

Horizons: $h \in \{1, 5, 21\}$ trading days

Target: Log returns $r_{t \rightarrow t+h} = \ln(P_{t+h}) - \ln(P_t)$

Features

- Technical: momentum, MA, volatility, RSI, MACD
- Earnings: surprise %, days to/from announcement
- Event windows (± 3 days)

Leakage Controls

- Forward-only validation
- 5-day embargos
- Scaling fit on train only

Data Sources

- Prices: yfinance
- Earnings: API Ninjas
- Macro: FRED API

Training

- Pooled across all 30 stocks
- Shared parameters

Tabular Models

- **Ridge Regression**
 - Linear baseline
 - L2 regularization
 - α tuned on validation
- **Random Forest**
 - 100 trees, depth 5
 - Strong tabular learner
- **XGBoost**
 - Gradient boosting
 - Depth 3, lr 0.1

Trained separately per horizon

Sequence Models

- LSTM (64 hidden, 60-day lookback)
- GRU (simpler than LSTM)
- TFT (4 attn heads, $\sim 50k$ params)

Predict all horizons simultaneously

Metrics: RMSE, R^2 , Directional Accuracy (DA)

- DA = fraction of correct directional predictions
- 50% = random; $>50\%$ = predictive value

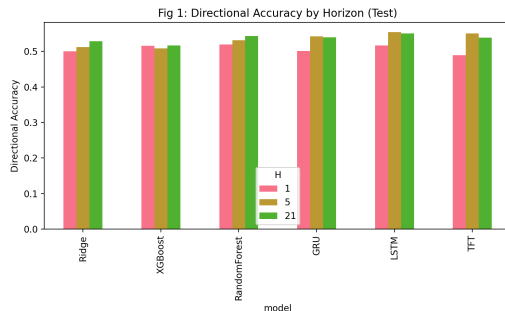
Main Results: Performance by Horizon

Model	h	RMSE	R^2	DA
<i>$h = 1$ (next day)</i>				
LSTM	1	0.0162	-0.003	51.7%
GRU	1	0.0162	-0.006	50.1%
Rand. Forest	1	0.0164	-0.023	51.9%
Ridge	1	0.0163	-0.018	50.0%
<i>$h = 5$ (one week)</i>				
Ridge	5	0.0365	-0.020	51.2%
Rand. Forest	5	0.0397	-0.207	53.1%
LSTM	5	0.0578	-0.002	55.4%
GRU	5	0.0579	-0.003	54.2%
<i>$h = 21$ (one month)</i>				
LSTM	21	0.0508	-0.002	55.0%
GRU	21	0.0509	-0.002	53.9%
TFT	21	0.0510	-0.007	53.9%
Ridge	21	0.0742	-0.028	52.8%

Key Takeaway

LSTM wins at $h = 1$ & $h = 21$. **Ridge** dominates at $h = 5$. TFT competitive but not superior.

Key Observations



- Sequence models (LSTM/GRU) improve with horizon
- At $h = 21$: LSTM reaches 55% DA
- TFT competitive but not superior
- Negative R^2 values normal in finance (low signal/noise)

H1: TFT Superiority?

At $h = 21$:

- LSTM: 55.0% DA
- TFT: 53.9% DA

NOT SUPPORTED

LSTM matches TFT; attention provides no advantage

H3: Regime Dependence?

Best models:

- Bear 2022: GRU
- Rally 2023–24: LSTM

SUPPORTED

Rankings change across regimes

H2: Earnings Features Matter

Ablation at $h = 21$

Features	DA
Tech only	52.2%
+Earnings	53.3%

Overall: +1.1 pp improvement

Earnings Windows

Period	DA
Earnings ($\pm 3d$)	61.4%
Non-earnings	53.9%

+7.5 pp during announcements!

Verdict: H2 SUPPORTED

Substantial improvements during earnings windows confirm post-earnings-announcement drift.

- 1 **Data constraints:** Small panel (30 stocks, daily)—limited for deep learning
- 2 **Low signal-to-noise:** Short-term patterns quasi-linear
- 3 **Model capacity:** TFT ($\sim 50k$ params) vs. LSTM ($\sim 25k$)—overfitting risk
- 4 **Feature engineering:** Technical indicators capture most signal

When do transformers help?

- Large datasets (100B+ time points)
- Multimodal inputs (text + prices)
- Pretraining / transfer learning

Main Findings

- 1 Simple models competitive (LSTM/Ridge)
- 2 Earnings features valuable (+7.5 pp DA)
- 3 Performance regime-dependent
- 4 Transformers need scale

Future Work

- Multimodal integration (text + prices)
- Larger scale (S&P 500, intraday)
- Advanced architectures (PatchTST)
- Economic evaluation

Bottom Line

Start simple, engineer features carefully, validate rigorously. Transformers excel with scale and multimodal data.

Thank You!

Questions?

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